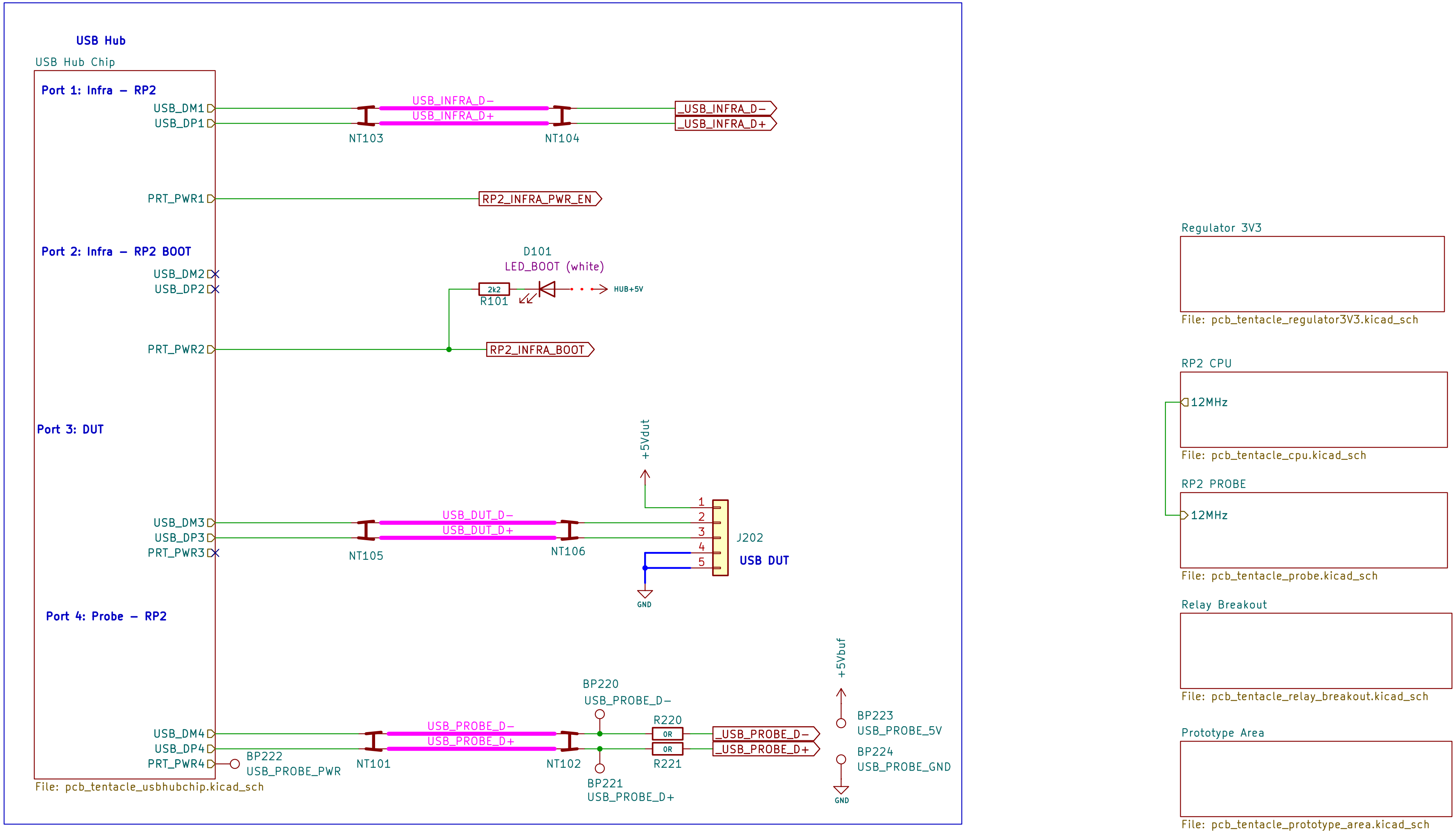

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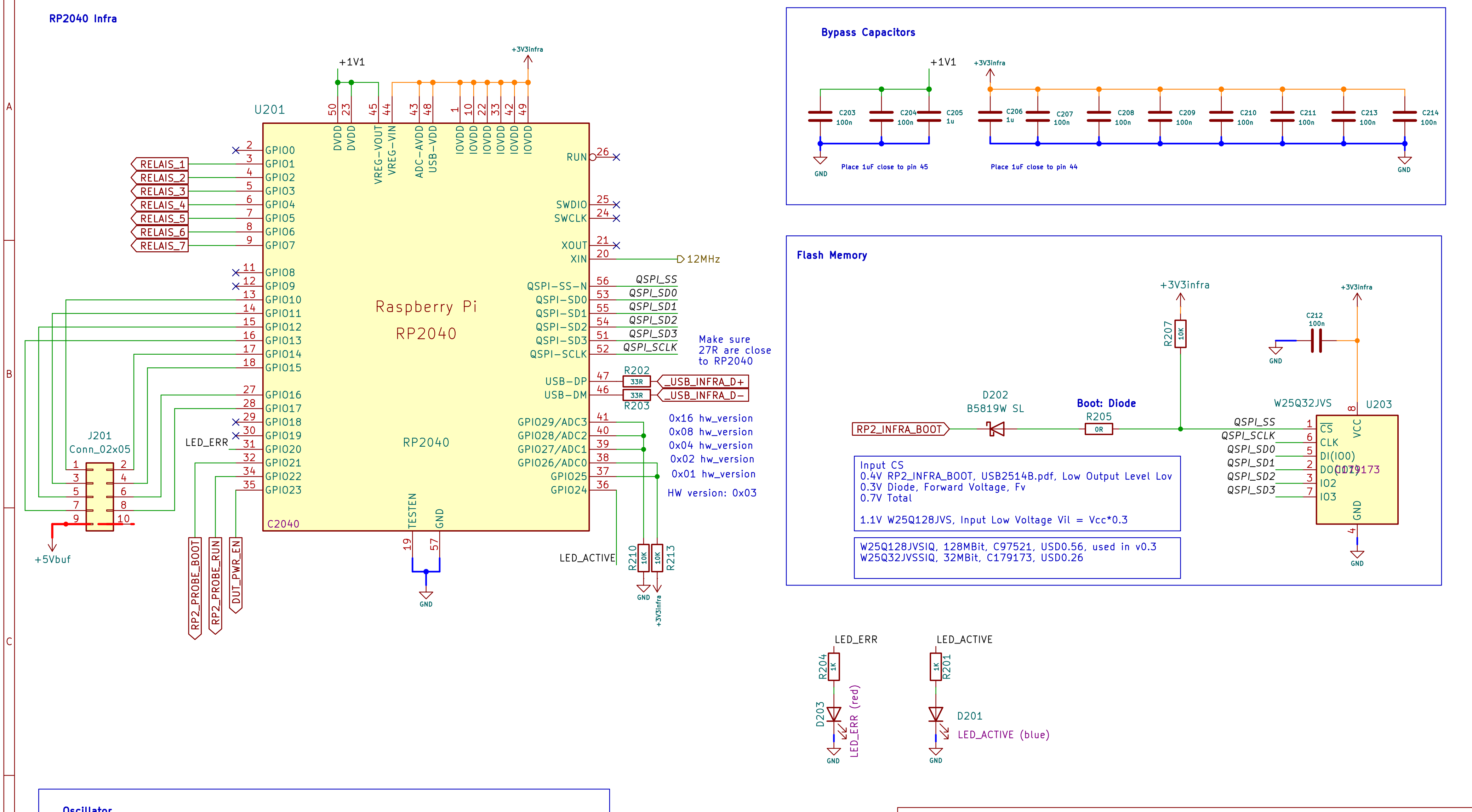
Sheet: /
File: pcb_tentacle.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-07-12	**Rev: 0.8.0**
KiCad E.D.A. 10.0.4		Id: 1/14



JLC alternatives

jlc	USD/Stock	Package	Comments
C2831468	0.37/1k	SMD2520-4P	1nd
C2901596	0.37/1k	SMD2520-4P	2nd
C437139	0.29/188	SMD2520-4P	2nd
C387449	0.49/3k	SMD2520-4P	2nd
C409301	0.55/2k	SMD2520-4P	2nd

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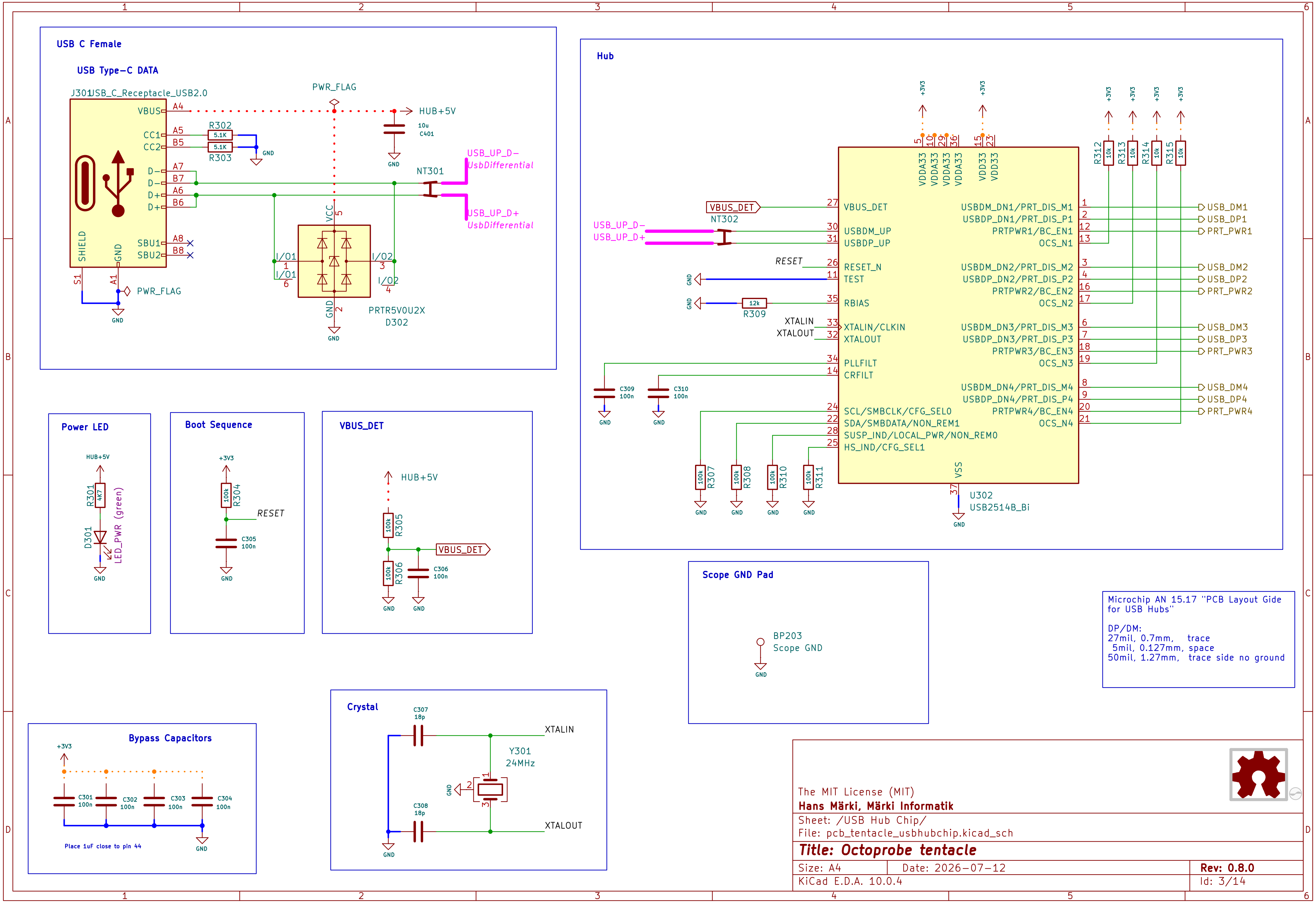
Sheet: /RP2 CPU/
File: pcb_tentacle_cpu.kicad_sch

Title: Octoprobe tentacle

Size: A4 Date: 2026-07-12 **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 Id: 2/14



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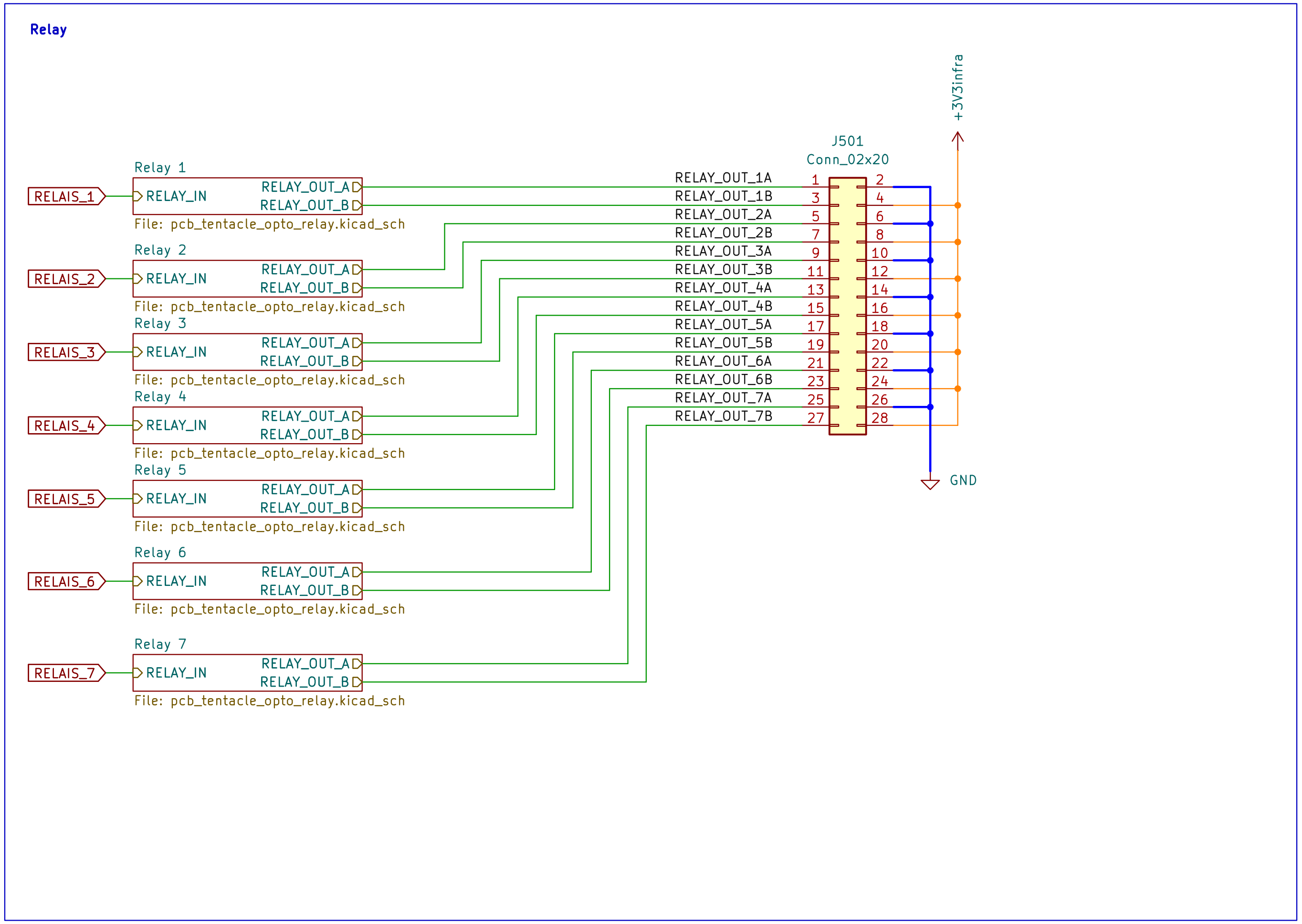
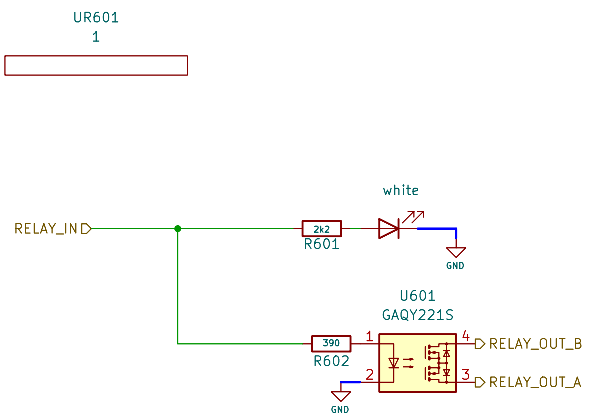
Sheet: /USB Hub Chip/
File: pcb_tentacle_usbhubchip.kicad_sch
Title: Octoprobe tentacle
Size: A4 Date: 2026-07-12 **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 Id: 3/14



Hans Märki, Märki Informatik
Sheet: /Relay Breakout/
File: pcb_tentacle_relay_breakout.kicad_sch

Title: Octoprobe tentacle

Size: A4 | Date: 2026-07-12 | **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 | Id: 5/14



Calculation of the input resistor

Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

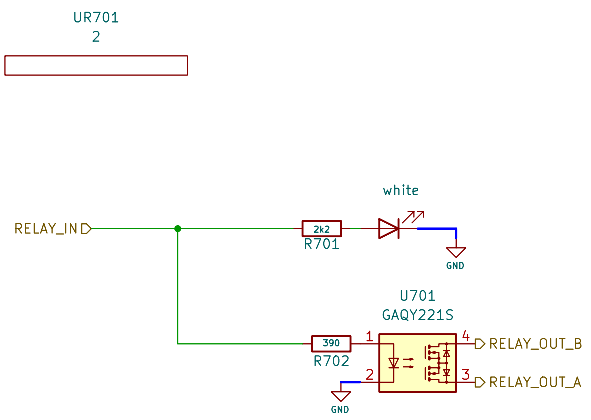
390 Ohm

Hans Märki, Märki Informatik

Sheet: /Relay Breakout/Relay 1/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-07-12	**Rev: 0.8.0**
KiCad E.D.A. 10.0.4		Id: 6/14



Calculation of the input resistor

Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

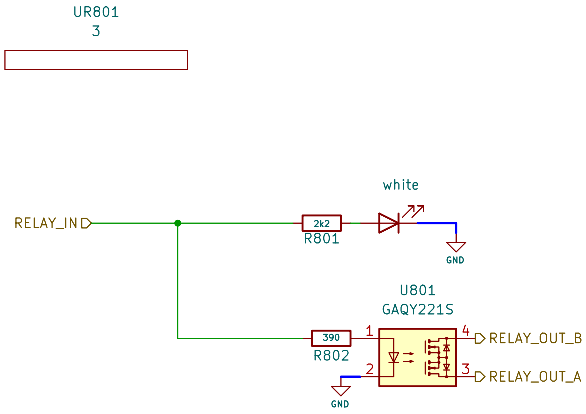
390 Ohm

Hans Märki, Märki Informatik

Sheet: /Relay Breakout/Relay 2/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-07-12	**Rev: 0.8.0**
KiCad E.D.A. 10.0.4		Id: 7/14



Calculation of the input resistor

Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

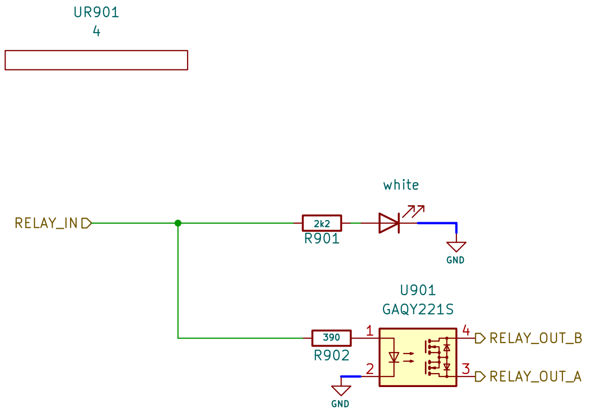
390 Ohm

Hans Märki, Märki Informatik

Sheet: /Relay Breakout/Relay 3/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-07-12	**Rev: 0.8.0**
KiCad E.D.A. 10.0.4		Id: 8/14



Calculation of the input resistor

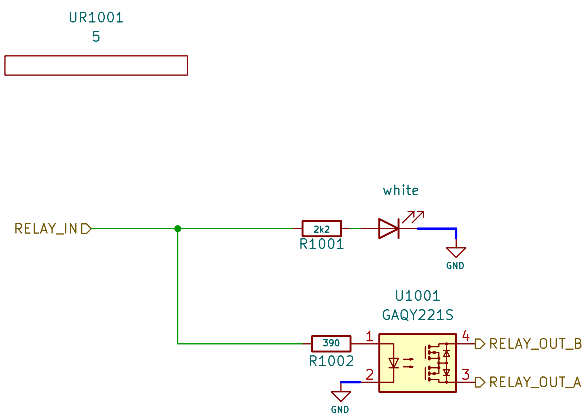
Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

390 Ohm

Hans Märki, Märki Informatik
Sheet: /Relay Breakout/Relay 4/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4 | Date: 2026-07-12 | **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 | Id: 9/14



Calculation of the input resistor

Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

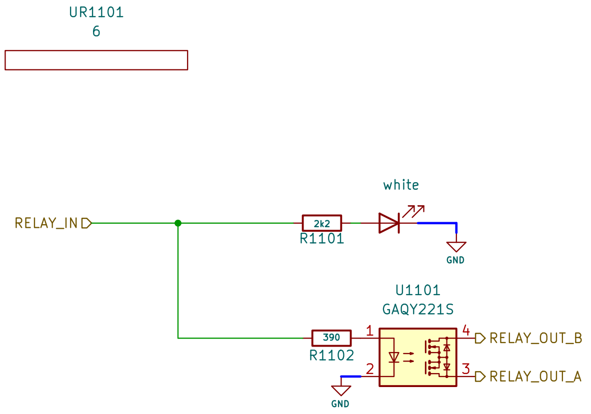
390 Ohm

Hans Märki, Märki Informatik

Sheet: /Relay Breakout/Relay 5/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4 | Date: 2026-07-12 | **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 | Id: 10/14



Calculation of the input resistor

Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

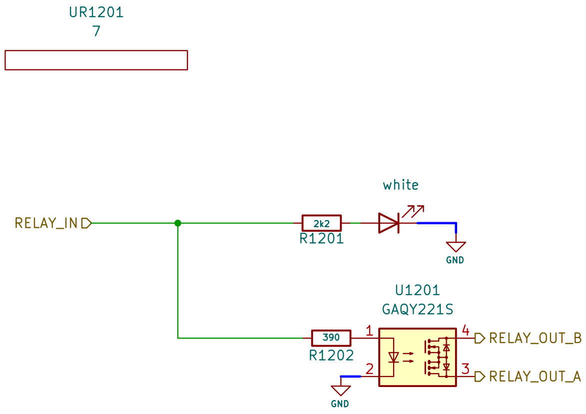
390 Ohm

Hans Märki, Märki Informatik

Sheet: /Relay Breakout/Relay 6/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4 | Date: 2026-07-12 | **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 | Id: 11/14



Calculation of the input resistor

Vdiff = 3.3V - Vf = 3.3V- 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

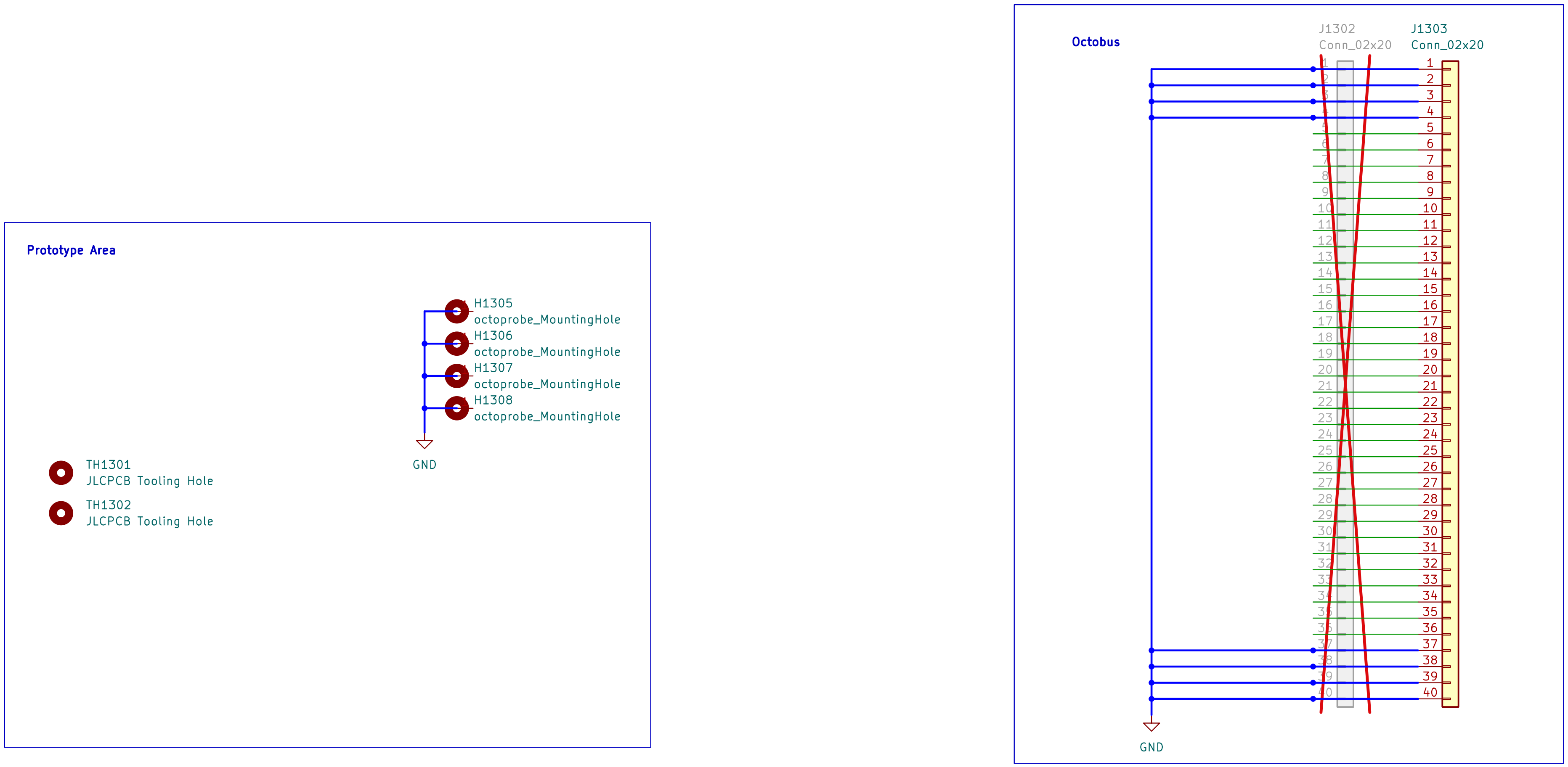
390 Ohm

Hans Märki, Märki Informatik

Sheet: /Relay Breakout/Relay 7/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4 | Date: 2026-07-12 | **Rev: 0.8.0**
KiCad E.D.A. 10.0.4 | Id: 12/14

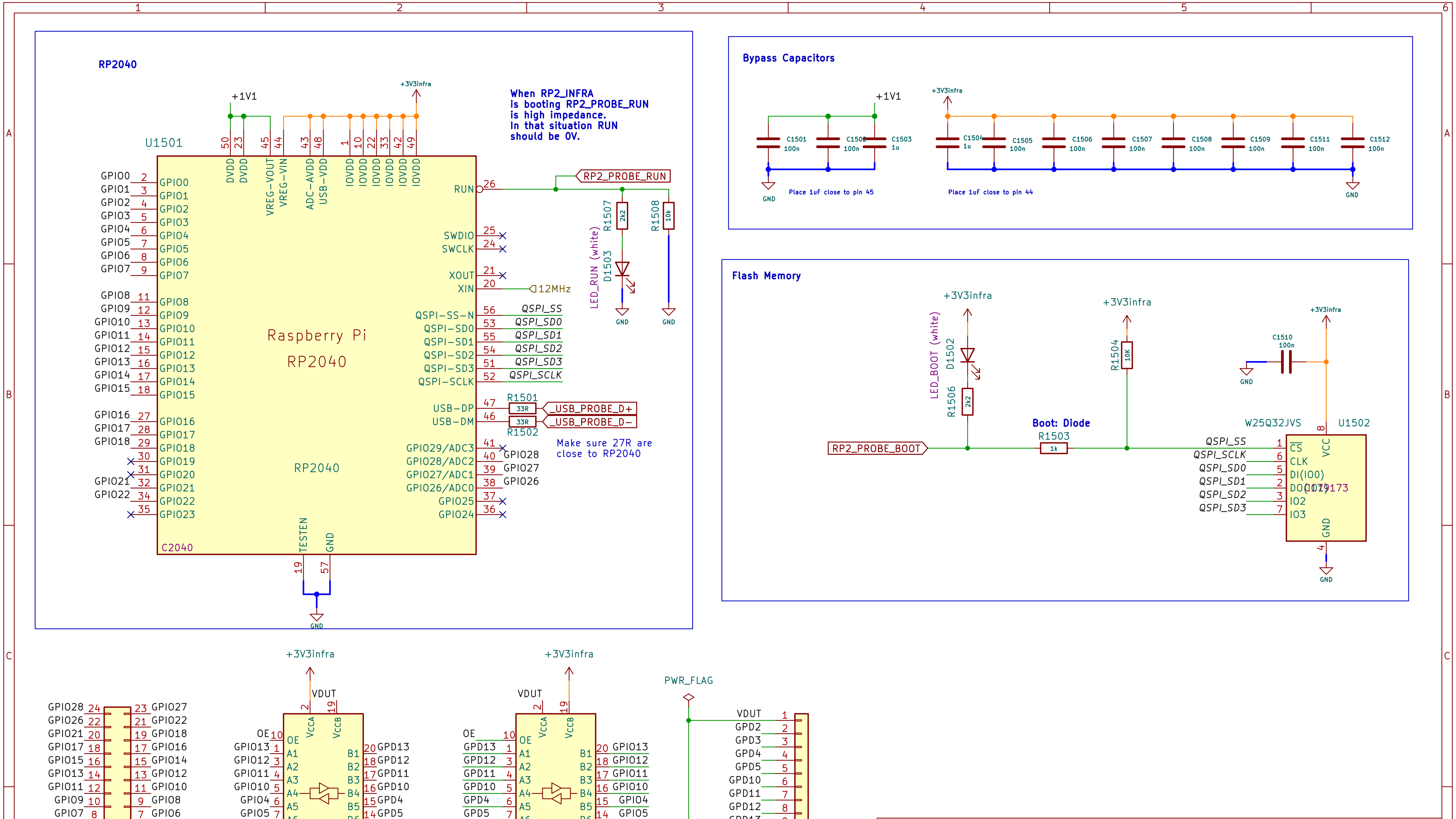


Hans Märki, Märki Informatik

Sheet: /Prototype Area/
File: pcb_tentacle_prototype_area.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-07-12	**Rev: 0.8.0**
KiCad E.D.A. 10.0.4		Id: 13/14



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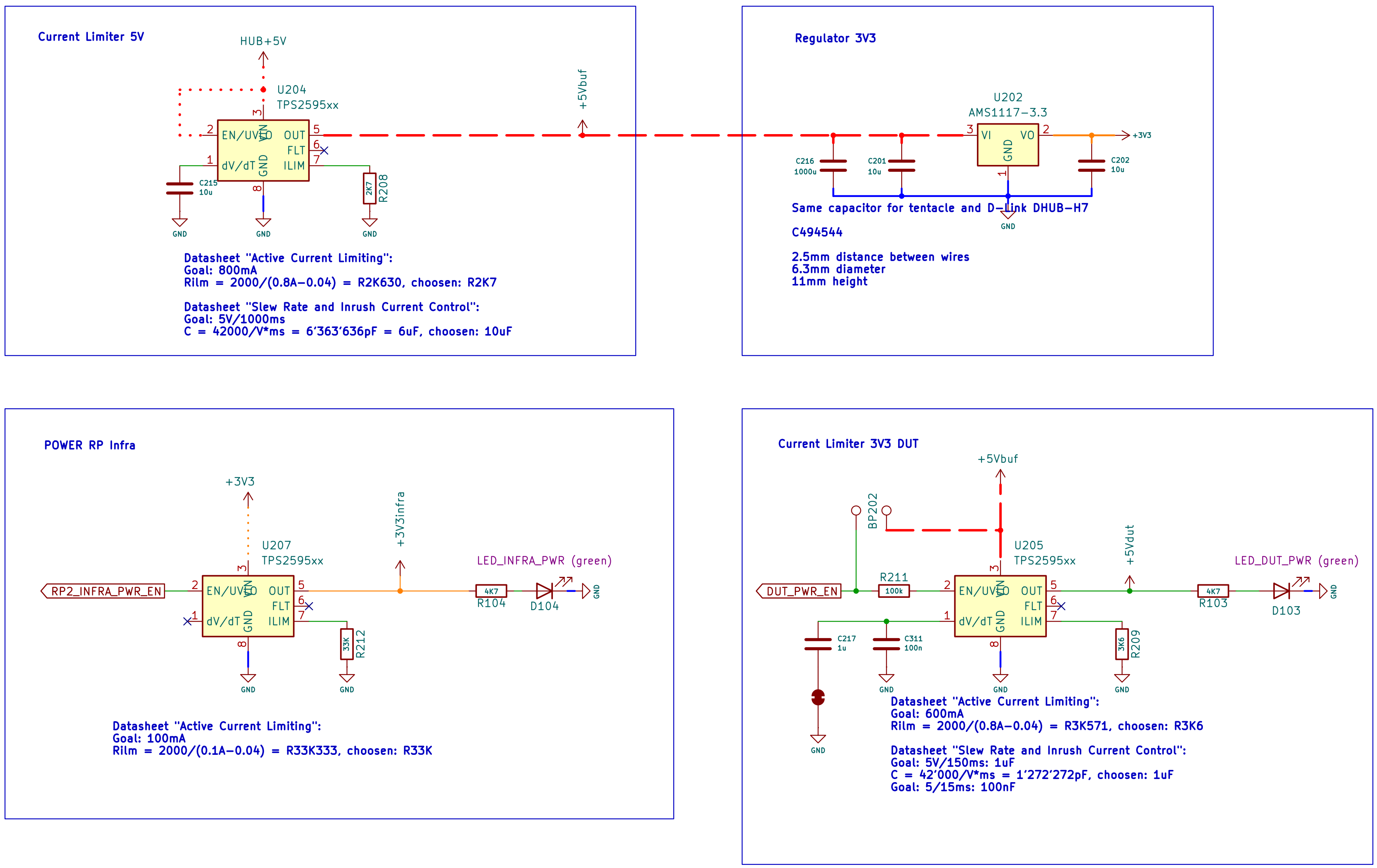
Sheet: /RP2 PROBE/

File: pcb_tentacle_probe.kicad_sch

Title: Octoprobe tentacle

Size: A4 Date: 2026-07-12 **Rev: 0.8.0**

KiCad E.D.A. 10.0.4 Id: 15/14



Sheet: /Regulator 3V3/
File: pcb_tentacle_regulator3V3.kicad_sch

Title:

Size: A4 | Date: | Rev:

KiCad E.D.A. 10.0.4 | Id: 16/14